Teaching notes and solutions

> **Teacher note · Teacher notes.** Distinguish the unknown function from the real variable. Have each antiderivative checked by differentiation. For $y' = ay + b$, look first for the constant solution. The case $a = 0$ must remain visible. The trigonometric antiderivatives are given here as a reference table, but their assessed use awaits the trigonometry unit. For Euler, keep the units and compare with a known exact solution.

Solutions to the activity

1. The equation is $T' = -0.1T + 2$. A constant solution is 20; all the solutions are $T(t) = 20 + Ce^{-0.1t}$. Since $T(0) = 80$, $C = 60$. We check $T'(t) = -6e^{-0.1t} = -0.1(T(t) - 20)$ and $T(0) = 80$.
2. $T'(t) = -6e^{-0.1t} < 0$: the temperature decreases strictly. Since $e^{-0.1t} \to 0$ at $+\infty$, $T(t) \to 20\,^\circ\text{C}$, while remaining above $20\,^\circ\text{C}$.
3. $T(t) = 40 \iff e^{-0.1t} = \frac{1}{3}$. Taking the logarithm, $-0.1t = -\ln 3$, that is $t = 10\ln 3 \simeq 10.9861$ min. Strict decrease ensures that this is the only instant at which the temperature passes through $40\,^\circ\text{C}$.
4. For $h = 1$, $Y_{k+1} = Y_k - 0.1(Y_k - 20)$. Thus $Y_1 = 80 - 6 = 74$ and $Y_2 = 74 - 5.4 = 68.6$. The exact solution gives $T(2) = 20 + 60e^{-0.2} \simeq 69.1238$. Euler underestimates the temperature here by about $0.5238\,^\circ\text{C}$.
5. The model assumes a room temperature of $20\,^\circ\text{C}$ and a constant exchange coefficient of $0.1\,\text{min}^{-1}$. A change of room, a lid or stirring can alter these assumptions; the formula is then no longer automatically appropriate.

E1. Linearity gives $F(x) = x^3 - 2x^2 + x + C$, $C \in \mathbb{R}$. We check $F'(x) = 3x^2 - 4x + 1$. All the antiderivatives on the interval $\mathbb{R}$ differ by a constant.

E2. The denominator is positive and its derivative is $2x$. The antiderivatives are $\ln(x^2+4) + C$. The condition at 0 gives $C = -\ln 4$, so $F(x) = \ln(x^2+4) - \ln 4$; we check $F(0) = 0$ and $F'(x) = \frac{2x}{x^2+4}$.

E3. The solutions of $y' = 3y$ are Ce^{3x}. The condition $Ce^3 = 2$ forces $C = 2e^{-3}$. Thus $y(x) = 2e^{3(x-1)}$, whose derivative is $3y(x)$ and whose value at 1 is 2.

E4. The constant 4 is a particular solution. The solutions are therefore $4 + Ce^{-x}$. Since $4 + C = 7$, $C = 3$. The solution is $y(x) = 4 + 3e^{-x}$; its derivative $-3e^{-x}$ is indeed $-y + 4$, and $y(0) = 7$.

E5. $g(x) = x + 1$ satisfies $g' = 1 = g - x$. If $z = y - g$, then $z' = z$, so $y = x + 1 + Ce^x$. The condition $y(0) = 0$ forces $C = -1$. We obtain $y = x + 1 - e^x$ and we check $y' = 1 - e^x = y - x$.

Solutions to HW7

1. If $g(t) = at + b$, then $g' = a$ and $-g + 2t + 1 = (2 - a)t + 1 - b$. For the identity to hold for every t, we need $2 - a = 0$ and $a = 1 - b$. So $a = 2$, $b = -1$, and $g(t) = 2t - 1$.
2. Let $z = y - g$. Subtracting the two equations, $z' = -z$, so $z = Ce^{-t}$. Thus $y(t) = 2t - 1 + Ce^{-t}$. Since $y(0) = 3$, $C = 4$. The function $y(t) = 2t - 1 + 4e^{-t}$ does satisfy $y' = 2 - 4e^{-t} = -y + 2t + 1$ and $y(0) = 3$.

3. $y'(t) = 2 - 4e^{-t}$. This derivative is negative for $t < \ln 2$, zero at $\ln 2$, positive afterwards. The solution decreases on $[0, \ln 2]$ then increases on $[\ln 2, +\infty[$. Its minimum is $y(\ln 2) = 2\ln 2 - 1 + 2 = 1 + 2\ln 2$.

4.
```
def euler_dm7(N):
    h = 2/N
    y = 3.0
    for k in range(N):
        t = k*h
        y = y+h*(-y+2*t+1)
    return y
```

   The time used in the right-hand side is that at the start of the step. The program gives $Y_{10} \simeq 3.42950$ and $Y_{100} \simeq 3.53048$, whereas $y(2) = 3 + 4e^{-2} \simeq 3.54134$. The absolute errors are about 0.11184 and 0.01086. To check these values without a loop, let $Z_k = Y_k - (2kh - 1)$. Euler's recurrence gives $Z_{k+1} = (1-h)Z_k$ and $Z_0 = 4$. So $Y_N = 3 + 4(1 - \frac{2}{N})^N$.

5. A solution of a differential equation must satisfy an equality between functions on the whole interval, as well as the initial condition. A few numerical values only check a finite number of points; the symbolic substitution of question 2 is a check on the whole domain.

Solutions to the textbook or past-paper work

Exercise 30. If y is constant equal to ℓ, the equation becomes $0 = -\frac{1}{2}\ell + 3$, so $\ell = 6$. Setting $z = y - 6$, we obtain $z' = -\frac{1}{2}z$; all the solutions are $y(x) = 6 + Ce^{-\frac{x}{2}}$, with $C \in \mathbb{R}$.

Exercise 25.

1. $f' = \frac{1}{2}f$ gives $f(x) = Ce^{\frac{x}{2}}$. Since $f(0) = 1$, $C = 1$.

2. $5f' = f$ gives $f(x) = Ce^{\frac{x}{5}}$. The condition $f(1) = 0$ forces $Ce^{\frac{1}{5}} = 0$; since the exponential is never zero, $C = 0$. The solution is the zero function.

3. $f' + f = 0$ gives $f(x) = Ce^{-x}$, so $f'(x) = -Ce^{-x}$. The condition concerns the derivative: $f'(1) = 2$ is equivalent to $-Ce^{-1} = 2$, that is $C = -2e$. Thus $f(x) = -2e^{1-x}$.